# LEONORA, Australia

WMO: **944480**

Lat: **28.884S** Lon: **121.330E** Elev: **380** StdP: **96.84** Time Zone: **8.00 (E08)** Period: **94-05** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **7** | **3.6** | **4.8** | **-3.6** | **2.9** | **18.5** | **-2.3** | **3.2** | **18.3** | **12.2** | **16.6** | **10.2** | **18.1** | **0.6** | **90** | **0.362** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **1** | **13.4** | **41.0** | **20.3** | **39.4** | **20.1** | **37.8** | **19.8** | **24.9** | **31.4** | **23.6** | **31.1** | **22.5** | **30.7** | **2.3** | **270** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **23.3** | **18.9** | **28.3** | **21.8** | **17.3** | **26.6** | **20.1** | **15.5** | **26.1** | **78.3** | **31.2** | **72.8** | **31.0** | **68.1** | **29.9** | **35.4** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **10.2** | **8.4** | **7.8** | DB | **1.0** | **43.6** | **0.8** | **1.3** | **0.4** | **44.5** | **-0.1** | **45.3** | **-0.5** | **46.1** | **-1.1** | **47.0** | (4) |
| (5) | | | | WB | **0.4** | **27.8** | **0.9** | **3.3** | **-0.3** | **30.1** | **-0.8** | **32.1** | **-1.3** | **33.9** | **-2.0** | **36.3** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **20.6** | **29.2** | **27.5** | **23.9** | **20.6** | **16.7** | **13.0** | **12.5** | **13.7** | **17.8** | **20.8** | **24.3** | **27.4** | (6) |
| (7) | | | DBStd | **6.74** | **3.95** | **4.16** | **4.32** | **3.08** | **3.63** | **2.71** | **2.72** | **3.34** | **3.56** | **3.58** | **4.03** | **4.14** | (7) |
| (8) | | | HDD10.0 | **13** | **0** | **0** | **0** | **0** | **0** | **4** | **7** | **2** | **0** | **0** | **0** | **0** | (8) |
| (9) | | | HDD18.3 | **660** | **0** | **0** | **4** | **13** | **77** | **162** | **182** | **150** | **53** | **16** | **2** | **1** | (9) |
| (10) | | | CDD10.0 | **3874** | **596** | **491** | **430** | **318** | **208** | **93** | **85** | **118** | **233** | **333** | **430** | **538** | (10) |
| (11) | | | CDD18.3 | **1478** | **337** | **258** | **176** | **82** | **27** | **1** | **1** | **8** | **36** | **91** | **182** | **281** | (11) |
| (12) | | | CDH23.3 | **18733** | **4880** | **3441** | **1989** | **696** | **206** | **8** | **9** | **84** | **455** | **1057** | **2162** | **3746** | (12) |
| (13) | | | CDH26.7 | **10216** | **3056** | **2013** | **1006** | **222** | **44** | **1** | **1** | **18** | **137** | **438** | **1094** | **2187** | (13) |
| (14) | Wind | | WSAvg | **2.1** | **2.4** | **2.3** | **2.2** | **1.5** | **1.5** | **1.6** | **1.7** | **1.9** | **2.3** | **2.7** | **2.8** | **2.7** | (14) |
| (15) | Precipitation | | PrecAvg | **249** | **19** | **42** | **27** | **21** | **27** | **26** | **23** | **16** | **10** | **13** | **14** | **19** | (15) |
| (16) | | | PrecMax | **552** | **128** | **285** | **179** | **101** | **110** | **100** | **102** | **79** | **47** | **56** | **52** | **94** | (16) |
| (17) | | | PrecMin | **94** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | (17) |
| (18) | | | PrecStd | **109** | **28** | **59** | **39** | **23** | **28** | **24** | **26** | **18** | **11** | **16** | **13** | **22** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **43.2** | **44.0** | **39.9** | **33.6** | **30.0** | **24.1** | **24.4** | **29.2** | **32.8** | **36.8** | **39.2** | **41.8** | (19) |
| (20) | | | MCWB | | **20.9** | **20.9** | **20.6** | **17.7** | **16.2** | **14.7** | **13.2** | **15.0** | **16.0** | **17.1** | **18.9** | **19.8** | (20) |
| (21) | | 2% | DB | | **41.2** | **40.7** | **37.3** | **31.2** | **27.6** | **22.0** | **22.4** | **25.6** | **30.1** | **34.2** | **36.7** | **39.6** | (21) |
| (22) | | | MCWB | | **20.5** | **20.4** | **19.8** | **17.4** | **15.8** | **13.4** | **12.4** | **13.7** | **15.1** | **17.6** | **18.6** | **19.4** | (22) |
| (23) | | 5% | DB | | **39.6** | **38.5** | **34.9** | **29.4** | **25.6** | **20.4** | **20.5** | **23.0** | **28.1** | **31.8** | **34.9** | **37.9** | (23) |
| (24) | | | MCWB | | **20.3** | **20.6** | **19.6** | **17.6** | **15.7** | **12.7** | **12.3** | **12.9** | **14.7** | **16.7** | **18.3** | **19.4** | (24) |
| (25) | | 10% | DB | | **37.8** | **36.1** | **32.3** | **27.5** | **23.4** | **18.8** | **18.7** | **20.8** | **25.9** | **29.3** | **32.7** | **36.0** | (25) |
| (26) | | | MCWB | | **20.1** | **20.4** | **19.5** | **17.1** | **15.0** | **12.6** | **11.8** | **12.0** | **14.1** | **15.5** | **17.6** | **19.3** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **24.4** | **26.5** | **25.3** | **22.5** | **19.2** | **17.5** | **16.3** | **16.1** | **17.6** | **28.5** | **22.2** | **26.9** | (27) |
| (28) | | | MCDB | | **34.9** | **33.0** | **30.5** | **26.9** | **24.6** | **20.2** | **19.3** | **24.3** | **26.2** | **29.3** | **31.4** | **33.4** | (28) |
| (29) | | 2% | WB | | **23.3** | **25.1** | **23.1** | **20.6** | **18.0** | **15.7** | **14.6** | **14.9** | **16.4** | **18.7** | **20.2** | **23.5** | (29) |
| (30) | | | MCDB | | **34.8** | **31.5** | **28.3** | **25.9** | **24.4** | **19.2** | **19.0** | **22.5** | **26.8** | **28.8** | **31.7** | **30.3** | (30) |
| (31) | | 5% | WB | | **22.4** | **24.1** | **21.7** | **19.4** | **17.0** | **14.6** | **13.4** | **13.6** | **15.5** | **17.1** | **19.3** | **21.5** | (31) |
| (32) | | | MCDB | | **33.8** | **31.0** | **28.3** | **24.9** | **22.2** | **18.2** | **18.2** | **21.5** | **25.8** | **28.5** | **31.4** | **32.2** | (32) |
| (33) | | 10% | WB | | **21.4** | **22.9** | **20.6** | **18.5** | **16.1** | **13.3** | **12.6** | **12.5** | **14.7** | **16.1** | **18.4** | **20.2** | (33) |
| (34) | | | MCDB | | **33.5** | **29.5** | **29.4** | **23.9** | **21.2** | **17.3** | **17.2** | **19.8** | **24.5** | **27.8** | **30.0** | **32.8** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **13.4** | **12.3** | **11.5** | **11.5** | **11.1** | **10.9** | **11.4** | **12.6** | **14.1** | **13.5** | **12.6** | **12.7** | (35) |
| (36) | | 5% DB | MCDBR | | **15.3** | **16.1** | **15.0** | **14.4** | **13.5** | **12.9** | **14.2** | **15.2** | **16.9** | **17.3** | **14.8** | **14.8** | (36) |
| (37) | | | MCWBR | | **4.7** | **5.1** | **5.3** | **5.6** | **5.7** | **6.3** | **7.1** | **7.3** | **6.7** | **7.0** | **5.2** | **5.4** | (37) |
| (38) | | 5% WB | MCDBR | | **13.5** | **11.5** | **11.7** | **10.9** | **10.3** | **9.8** | **10.8** | **13.0** | **14.8** | **14.3** | **13.5** | **13.3** | (38) |
| (39) | | | MCWBR | | **4.9** | **4.5** | **6.0** | **5.3** | **4.9** | **6.0** | **6.8** | **7.0** | **6.4** | **6.9** | **5.1** | **6.6** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.341** | **0.344** | **0.324** | **0.311** | **0.289** | **0.278** | **0.276** | **0.283** | **0.300** | **0.321** | **0.328** | **0.330** | (40) |
| (41) | | taud | | | **2.580** | **2.591** | **2.622** | **2.630** | **2.654** | **2.666** | **2.661** | **2.619** | **2.557** | **2.514** | **2.530** | **2.568** | (41) |
| (42) | | Ebn at Noon | | | **1000** | **980** | **969** | **935** | **914** | **904** | **921** | **953** | **978** | **992** | **1006** | **1013** | (42) |
| (43) | | Edh at Noon | | | **106** | **102** | **93** | **84** | **74** | **69** | **72** | **83** | **98** | **109** | **111** | **108** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **7.79** | **6.93** | **6.01** | **4.87** | **3.92** | **3.31** | **3.65** | **4.71** | **6.08** | **7.07** | **7.82** | **8.19** | (44) |
| (45) | | RadStd | | | **0.62** | **0.57** | **0.43** | **0.26** | **0.25** | **0.21** | **0.24** | **0.26** | **0.24** | **0.44** | **0.34** | **0.46** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (3 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page